WHITE PAPER

Healthcare Reimagined:

The Infrastructure Layer for Cash-Based, Direct-to-Consumer Healthcare

Connecting Diagnostic Laboratories, Telehealth Providers, Collection Sites, Imaging Centers, Direct Primary Care Practices, and Independent Pharmacies Through a Unified Platform

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Executive Summary

The American healthcare system spends approximately **$4.5 trillion annually** — nearly 18% of GDP — yet 25–30 million Americans remain uninsured, tens of millions more are underinsured, and the system consistently underperforms on access, outcomes, and patient satisfaction. At the heart of this paradox lies a delivery model burdened by administrative complexity, insurance intermediaries, opaque pricing, and deeply fragmented care.

This white paper makes the case that the solution is not another vertically integrated healthcare company attempting to own every layer of the stack. The solution is **connectivity infrastructure** — a technology platform that enables the thousands of independent laboratories, telehealth providers, collection sites, and imaging centers already operating in the market to function as a unified, cash-based healthcare ecosystem without surrendering their independence or their brands.

The platform model does not require building a single collection site, hiring a single clinician, or negotiating a single lab contract. It requires building the **connective tissue** that allows existing healthcare participants to interoperate seamlessly — sharing orders, routing specimens, delivering results, and processing payments through a single technology layer that each participant accesses under their own brand.

This paper examines the systemic failures driving demand for this model, quantifies the market opportunity, documents the consumer preferences that make adoption inevitable, and describes the platform architecture that enables it. It is intended for healthcare entrepreneurs, investors, health system administrators, and policymakers who recognize that the industry's infrastructure deficit is as significant as its clinical challenges.

1. The Broken Architecture of American Healthcare

1.1 The Insurance Burden

The United States spends approximately $4.5 trillion annually on healthcare, yet **25–30 million Americans remain uninsured**, and tens of millions more are underinsured — meaning their coverage is practically unusable due to high deductibles, narrow networks, and prior authorization delays.

The average American family with employer-sponsored insurance now pays over **$6,000 per year in out-of-pocket costs** after premiums — a figure that has risen faster than wages for two consecutive decades. For a growing segment of the population, insurance has become a financial product that provides peace of mind rather than actual access to care.

1.2 Administrative Bloat and Price Opacity

Nearly **one-third of every dollar** spent on U.S. healthcare goes toward billing, coding, claims processing, and administrative overhead — not patient care. A single primary care visit may generate 15 or more administrative touchpoints between the provider, the insurer, and the patient before a claim is adjudicated.

Compounding this inefficiency is a near-total absence of price transparency. The same blood test may cost $8 at a cash-pay lab and $400 when billed through insurance, with the actual negotiated rate buried in proprietary contracts. Patients — treated as passive consumers rather than active decision-makers — have virtually no ability to shop for value.

1.3 Fragmented Care and the Referral Maze

The traditional care model forces patients through a labyrinth of referrals, specialist handoffs, and disconnected electronic health systems. A patient presenting with fatigue may wait weeks for a primary care appointment, receive a referral that takes additional weeks to fulfill, undergo laboratory testing whose results are communicated through a portal the patient struggles to access, and finally receive a follow-up appointment that may occur a month later.

This fragmentation is not merely inconvenient — it is **clinically dangerous**. Delayed diagnoses, medication errors at transitions of care, and loss-to-follow-up are among the leading drivers of preventable morbidity and mortality in the United States.

1.4 The Infrastructure Deficit No One Talks About

Beyond the well-documented failures of cost, access, and fragmentation lies a less visible but equally consequential problem: **the participants who could deliver better care lack the technology to coordinate with each other**. Independent laboratories process specimens but cannot receive electronic orders from telehealth platforms. Collection sites schedule appointments manually but have no digital connection to the labs that will process the tests. Telehealth providers order diagnostics but have no visibility into where patients can conveniently provide specimens or when results will arrive.

Each of these participants — laboratories, telehealth providers, collection sites, imaging centers — operates competently within its own silo. The system fails at the seams. And the seams are where patients fall through.

This is not a clinical problem. It is an **infrastructure problem**. And it requires an infrastructure solution.

2. The Case for Cash-Based, Direct-to-Consumer Healthcare

2.1 Defining the Model

A cash-based DTC healthcare system operates on a simple principle: patients pay directly for services at transparent, published prices, without routing payment through an insurance intermediary. This does not mean healthcare becomes unaffordable — quite the opposite. By eliminating the administrative overhead of insurance billing, providers can offer dramatically lower prices while maintaining or improving margins.

The DTC model is not new. Concierge medicine, direct primary care, and subscription health clinics have demonstrated its viability. What is new — and what this paper advocates — is the systematic integration of six critical components into a unified, technology-enabled ecosystem:

- **Diagnostic Laboratory Services** — offering comprehensive testing panels at transparent cash prices
- **Telehealth Platforms** — providing asynchronous and synchronous clinical consultations without insurance gatekeeping
- **Patient Collection Sites** — decentralized, accessible physical locations for specimen collection, vitals, and point-of-care testing
- **Imaging Centers** — freestanding outpatient facilities offering cash-pay X-ray, ultrasound, CT, and MRI at 30–70% below hospital pricing

Direct Primary Care Practices — membership-based physician practices offering transparent, insurance-free primary care with integrated diagnostic ordering at wholesale cash prices

Independent Pharmacies — community-based pharmacies extending beyond dispensing into diagnostic wellness testing, leveraging trusted patient relationships and high foot traffic to drive lab test adoption at the counter

2.2 Why Now? Market Conditions Favor Disruption

Several converging forces have created a historic window for this model to achieve mainstream adoption:

- **Post-Pandemic Telehealth Normalization:** Over 60% of Americans used telehealth at least once between 2020 and 2023, and 76% reported satisfaction equivalent to or better than in-person visits.
- **Rising Deductibles and HDHP Proliferation:** High-deductible health plans now cover 55% of insured workers, meaning the first $1,500–$7,000 of care annually is effectively self-pay regardless of insurance status.
- **Consumer Price Consciousness:** Price comparison has become culturally normalized. Patients who comparison-shop for airline tickets and mortgage rates increasingly expect the same from healthcare.
- **Lab Technology Democratization:** Advances in microfluidics, point-of-care diagnostics, and laboratory automation have dramatically reduced the cost of high-quality testing. A comprehensive metabolic panel can be processed for under $3 in variable cost at scale.
- **Regulatory Tailwinds:** CLIA waivers, expanded telehealth prescribing authority, and interoperability mandates under the 21st Century Cures Act collectively lower the barrier to integrated DTC care delivery.

- **DPC Movement Acceleration:** Over 2,500 direct primary care practices now operate in the United States, growing at 15–20% annually as physicians exit fee-for-service medicine. These membership-based practices operate entirely outside the insurance system and actively seek wholesale diagnostic pricing — making them natural platform participants with pre-qualified, cash-pay patient panels.
- **Independent Pharmacy Revenue Diversification:** PBM reimbursement compression has reduced independent pharmacy dispensing margins to 2–5%, forcing over 21,000 independent operators to seek new revenue streams. Diagnostic wellness testing requires no physical buildout or additional licensing and leverages the pharmacist's existing patient relationships — the highest-trust, highest-frequency healthcare touchpoint in American communities.

3. American Consumer Preferences and the Demand for Convenient Healthcare

No business case for healthcare transformation is complete without a deep examination of demand. The case for a connectivity platform is not merely a supply-side argument about more efficient delivery economics. It is equally, and perhaps more powerfully, a **demand-side story**: Americans are actively signaling, through their behavior, their spending, and their survey responses, that the traditional healthcare engagement model no longer fits their lives.

3.1 The Convenience Imperative

Americans do not primarily avoid healthcare because they distrust medicine or dislike physicians. They avoid it because accessing healthcare is, in practical terms, deeply inconvenient. A 2024 Accenture Health survey found that **61% of Americans have delayed or skipped a recommended healthcare service** in the past 12 months, with “too time-consuming” and “too difficult to schedule” ranking as the top two reasons — ahead of cost.

3.2 The Amazon Effect

The healthcare industry does not exist in a consumer experience vacuum. Americans’ tolerance for friction in any service category has been permanently recalibrated by two decades of digital consumer experience. Amazon delivers packages in two hours. DoorDash delivers restaurant meals in thirty minutes. Uber provides real-time pricing and a driver arrival time before a commitment is made. Against this backdrop, being told that the next available primary care appointment is in three to four weeks — for a non-urgent concern that will cost an unknown amount — is increasingly incomprehensible.

Patients do not evaluate healthcare access in isolation — they evaluate it against the best service experience they have had in any industry. The implicit benchmark is always the most convenient alternative available, not the most convenient healthcare alternative available.

3.3 The Rise of the Empowered Health Consumer

Wearable health technology is now used by **over 30% of American adults**. Home diagnostic testing — including COVID-19 rapid tests, at-home fertility panels, and direct-to-consumer genetic testing — has been adopted by tens of millions. Health and wellness apps collectively attract over **350 million American users annually**.

This population is not waiting for a physician to initiate a diagnostic conversation. They are proactively generating health data, interpreting it through consumer apps and online resources, and arriving at clinical encounters with specific hypotheses and questions. A connectivity platform that enables instant access to lab ordering, transparent pricing, and convenient collection is architecturally aligned with this behavior.

3.4 Generational Dynamics

Millennials and Gen Z now collectively represent the largest demographic cohort in the United States. A 2024 MGMA survey found that **45% of Millennials do not have a primary care physician** — not because they lack access, but because they view the traditional primary care model as incompatible with their lives. They are not anti-healthcare; they are anti-friction. They

will engage with care that is available on their schedule, priced transparently, and delivered through channels they already use.

3.5 Location and Access

The explosive growth of retail health — CVS MinuteClinic, Walgreens Health, and Walmart Health — demonstrated that when clinical services are placed in locations Americans already frequent, utilization rates increase dramatically. CVS MinuteClinic reported over **35 million patient visits** cumulatively by 2023, with satisfaction scores consistently exceeding traditional urgent care.

The lesson is not that retail health found the right model — the economic challenges at Walgreens and Walmart suggest it did not — but that location-based convenience is a genuine driver of utilization. A platform that connects thousands of existing collection sites into a unified network achieves the same geographic density without the capital burden of operating physical locations.

3.6 The Price Transparency Preference

A 2023 TransUnion Healthcare consumer survey found that 68% of patients said an unexpected medical bill negatively affected their perception of the provider, and 49% reported they would avoid seeking care in the future due to billing uncertainty. **82% of adults** in a 2024 Gallup healthcare poll identified knowing the cost upfront as the single most important factor in their healthcare purchase decision.

The cash-based model makes price transparency a core product feature. Every interaction begins with a published price. The patient knows before the encounter what the blood panel, telehealth consultation, and collection visit will cost in total. This is a fundamental reorientation of the provider-patient relationship from one defined by information asymmetry to one defined by mutual transparency.

4. The Platform Architecture: Connectivity, Not Ownership

The conventional approach to healthcare disruption is vertical integration — build the labs, hire the clinicians, open the sites, own the patient relationship. This approach requires enormous capital, carries significant regulatory risk, and forces the company to develop operational competence across multiple healthcare disciplines simultaneously. Walmart spent billions and exited. Walgreens Health continues to struggle. The unit economics of **operating** healthcare facilities at scale are punishing.

The platform model inverts this logic. Rather than building a healthcare company that needs technology, it builds a **technology company that healthcare companies need**. The platform does not own laboratories, employ clinicians, or operate collection sites. It provides the digital infrastructure that allows these independent participants to function as a coordinated ecosystem.

4.1 The Six-Sided Network

The platform connects six distinct participant categories, each of which derives independent value from connectivity while generating compounding value for the others:

Collection Sites: The Access Layer

Over 30,000 laboratory collection sites operate across the United States, ranging from hospital-affiliated draw stations to independent phlebotomy clinics, occupational health offices, and franchise networks. Most operate on locally managed scheduling systems with no digital integration to the laboratories that process their specimens or the telehealth providers who order the tests.

The platform provides collection sites with an e-commerce ordering front-end, automated scheduling, specimen tracking with unique IDs, integration with reference laboratories via HL7 messaging, and a branded patient portal for result delivery. Sites retain their brand, their pricing authority, and their patient relationships. The platform provides the operational infrastructure they lack.

Telehealth Providers: The Clinical Intelligence Layer

Telehealth serves as the clinical brain of the ecosystem — pre-test consultation to ensure appropriate test selection, results interpretation to convert laboratory data into actionable clinical guidance, and ongoing care management including prescription issuance and specialist referral.

The platform connects telehealth providers to the full collection site and laboratory network, enabling them to order diagnostics, route patients to the nearest collection point, receive results electronically, and manage longitudinal patient records. An asynchronous-first architecture reduces the cost per clinical encounter from $75–$150 (synchronous video) to $25–$45 (asynchronous review) while achieving equivalent clinical outcomes for the majority of primary care presentations.

High-Complexity Laboratories: The Diagnostic Engine

Reference laboratories process the specimens but often lack consumer-facing ordering and scheduling infrastructure. The platform provides a **white-label technology stack** that enables laboratories to offer direct-to-consumer ordering, automated scheduling at collection sites within their network, prescription authority integration in states that require it, and branded patient result portals — all without building their own consumer technology.

Laboratory testing is the single highest-leverage intervention in primary care. Studies consistently show that 70% of clinical decisions are informed by laboratory data, yet lab testing accounts for only 3–5% of total healthcare spend. This asymmetry makes laboratory services the ideal anchor for a cash-based healthcare ecosystem.

Imaging Centers: The Diagnostic Extension

Over **6,000 freestanding outpatient imaging centers** operate in the United States, offering cash-pay pricing 30–70% below hospital-based alternatives. A cash-pay MRI that costs $1,200–$3,000 at a hospital can be obtained for $250–$650 at an independent imaging center. The $19 billion annual outpatient imaging market represents a natural extension of the diagnostic ecosystem.

The platform connects imaging centers to the telehealth and collection site network, routing referrals from clinicians who need diagnostic imaging for their patients. Centers gain patient volume from the ecosystem; patients gain transparent pricing and convenient scheduling. The platform takes a revenue share on referrals it routes, creating alignment between volume generation and platform economics.

Direct Primary Care Practices: The Membership Medicine Layer

The direct primary care (DPC) movement has grown to over 2,500 practices across the United States, each operating on a membership model that eliminates insurance billing entirely. DPC physicians charge patients a flat monthly fee — typically $50–$150 per month — for unlimited primary care access, and avoid the administrative overhead that consumes 30–40% of revenue in traditional fee-for-service practices.

DPC practices face a persistent operational gap: their model promises transparent, affordable care, but diagnostic ordering remains tethered to legacy infrastructure. Most DPC physicians order labs through Quest or Labcorp portals designed for insurance-billed practices, resulting in pricing that contradicts the transparent, low-cost promise at the heart of the DPC value proposition. A DPC member paying $100/month for unlimited primary care may then receive a $400 bill for routine bloodwork ordered through a conventional laboratory channel.

The platform solves this by connecting DPC practices to the full laboratory and imaging ecosystem at wholesale cash prices. A comprehensive metabolic panel ordered through the platform costs $25–$45 instead of $200–$400 through insurance-billed channels. Imaging referrals route to freestanding centers at $250–$650 for an MRI instead of $1,200–$3,000 at a hospital. DPC physicians gain a unified ordering dashboard, automated patient routing to the nearest collection site from a planned network of 3,000+ collection sites accessible through channel partnerships, and electronic result delivery — all under pricing that aligns with their membership model.

For the platform, DPC practices represent a high-value participant category. Each practice brings a captive patient panel of 400–500 members who order an average of 4–4.5 lab tests per year and generate imaging referrals at a rate of 15–18% of patients annually. Because DPC members are already conditioned to pay cash for healthcare, conversion rates are substantially higher than general consumer marketing. The platform charges a $500/month access fee and 9% revenue share on lab and imaging GMV, with 70–75% of lab orders routed to collection sites within the network — feeding the collection fee flywheel that amplifies platform economics across all verticals.

The DPC market is growing at approximately 15–20% annually, driven by physician burnout in fee-for-service medicine and patient demand for relationship-based primary care. The American Academy of Family Physicians estimates that over 25% of family medicine residents now

express interest in DPC as a career path. Importantly, DPC practices skew toward younger, tech-comfortable physicians who are actively seeking digital tools to differentiate their practices — making them highly receptive to platform adoption. The typical DPC physician manages a panel of 400–600 patients (versus 2,000–2,500 in traditional primary care), enabling longer appointments, proactive screening, and a significantly higher per-patient diagnostic ordering rate.

The regulatory alignment between DPC and the platform is particularly strong. Because DPC practices operate entirely outside the insurance system, there is no claims adjudication, no prior authorization, and no network adequacy requirement to navigate. The DPC physician orders a test; the patient pays a transparent cash price; the platform routes the order. This eliminates the two highest-friction barriers in conventional diagnostic ordering — insurance verification and prior authorization — and creates the fastest order-to-result cycle in the ecosystem. In states with direct access testing laws, DPC patients can even self-order routine panels through the platform's e-commerce interface, with results flowing back to their DPC physician for clinical context.

Consider the DPC patient journey through the platform: A 42-year-old DPC member messages her physician about fatigue and weight gain. The physician orders a comprehensive thyroid panel, metabolic panel, and CBC through the platform's ordering dashboard — total patient cost: $85 at published cash prices. The platform routes the patient to a collection site three miles from her home with a next-day appointment. Specimens are collected, tracked via unique ID, and routed to the reference laboratory via HL7 integration. Results arrive in the patient portal within 48 hours and simultaneously populate the DPC physician's dashboard. The physician reviews results asynchronously, identifies subclinical hypothyroidism, and initiates treatment during the patient's next in-office visit. Total elapsed time from symptom report to diagnosis: 72 hours. Total cost: $85. Through a traditional insured pathway, the same sequence would involve a referral to endocrinology (3–6 week wait), insurance pre-authorization for the lab panel, a bill that arrives weeks later for an unpredictable amount, and results delivered through a hospital portal the patient may never check.

The strategic value of the DPC channel extends beyond direct revenue. DPC physicians are opinion leaders in the cash-pay healthcare movement. A platform that earns the loyalty of the DPC community gains organic advocacy among the most vocal proponents of transparent, patient-centered care — physicians who speak at conferences, publish newsletters, host podcasts, and actively recruit patients away from the insurance-dependent model. This word-of-mouth channel is effectively zero-cost customer acquisition for the platform.

Independent Pharmacies: The Community Access Layer

Over 21,000 independent pharmacies operate across the United States, serving as trusted healthcare touchpoints in communities where access to other clinical services may be limited. These pharmacies face an existential economic challenge: PBM reimbursement rates have compressed dispensing margins to the point where prescription revenue alone cannot sustain many independent operations. The average independent pharmacy's net profit margin has declined to 1–3%, with more than 90% of revenue concentrated in prescription dispensing — a dangerous single-source dependency.

The platform transforms independent pharmacies from prescription dispensaries into community diagnostic access points. Pharmacists — who already counsel patients on medications, manage chronic conditions, and provide immunizations — can now recommend wellness panels, medication therapy monitoring tests, and condition-specific diagnostics at the counter. The patient orders through a branded interface, the platform routes specimen collection to the nearest draw station (no in-pharmacy phlebotomy is required, eliminating CLIA concerns

entirely), and results are delivered electronically to the patient and, where appropriate, to the prescribing pharmacist for medication therapy management.

The economics are compelling on both sides. The pharmacy gains a new revenue stream at the lowest platform entry point ($200/month access fee, 9% revenue share on lab GMV, no setup fee), while the real value is indirect: lab findings identifying new conditions drive an average of 2.3 incremental prescription fills per finding — flowing directly back to the pharmacy's core dispensing business. For the platform, pharmacies represent massive geographic reach. A network of 1,000 pharmacies provides consumer access points in communities that may have no other connection to the diagnostic ecosystem, and 100% of pharmacy-originated lab orders route to collection sites, making pharmacies the single most efficient driver of the collection fee flywheel.

Critically, the pharmacy channel requires no clinical infrastructure beyond what already exists. Pharmacists do not perform phlebotomy, do not interpret results, and do not prescribe. They recommend testing based on clinical knowledge they already possess, and the platform handles the entire downstream workflow. This makes pharmacies the fastest vertical to activate and the easiest to scale.

The independent pharmacy market is undergoing a structural transformation that makes platform adoption not merely attractive but, for many operators, essential to survival. PBM-driven reimbursement compression has reduced average dispensing margins from 22–25% a decade ago to 2–5% today. DIR (Direct and Indirect Remuneration) fees — retroactive clawbacks imposed by PBMs after the point of sale — further erode profitability, with the National Community Pharmacists Association reporting that DIR fees exceeded $9 billion annually by 2023. Independent pharmacies are closing at a rate of approximately 1,000 per year, and the survivors are actively seeking revenue diversification. Diagnostic wellness testing through the platform represents a high-margin, high-frequency revenue stream that requires no additional staffing, no physical buildout, and no regulatory permitting beyond what the pharmacy already holds.

The pharmacy's competitive advantage as a platform participant is rooted in two structural factors that no other healthcare touchpoint replicates: frequency and trust. The average American visits a pharmacy 12–14 times per year — more than any other healthcare setting including primary care. Each visit is a potential conversion event for diagnostic testing. A pharmacist dispensing a statin can recommend a lipid panel. A pharmacist filling a thyroid medication can suggest a thyroid function test to monitor efficacy. A pharmacist counseling a patient on a new diabetes diagnosis can recommend a comprehensive metabolic panel and HbA1c. These are not cold sales pitches — they are clinically appropriate recommendations delivered by a trusted healthcare professional in the context of an existing care relationship. Studies consistently show that pharmacists rank among the most trusted healthcare professionals in the United States, with Gallup polling placing them in the top three for over two decades.

The regulatory framework for pharmacy-based diagnostic recommendations is well-established and favorable. Pharmacists in all 50 states are authorized to recommend over-the-counter diagnostic tests, and a growing number of states have expanded pharmacist scope of practice to include CLIA-waived point-of-care testing, collaborative practice agreements with physicians, and independent prescribing authority for select conditions. The platform's model avoids even these expanded-scope considerations entirely: the pharmacist recommends the test, the patient orders through the platform, and specimen collection occurs at a separate draw station. No in-pharmacy phlebotomy occurs, no CLIA waiver is required at the pharmacy location, and no diagnostic result interpretation is performed by the pharmacist. The pharmacy operates purely

as a recommendation and access channel, with the clinical and laboratory workflow managed entirely by the platform ecosystem.

Consider the pharmacy patient journey: A 58-year-old man picks up his metformin refill at his neighborhood pharmacy. The pharmacist, reviewing the dispensing record, notes that no HbA1c has been ordered in over nine months. She recommends a diabetes monitoring panel through the platform — HbA1c, comprehensive metabolic panel, and lipid panel — at a published cash price of $65. The patient orders on his phone via the pharmacy's branded interface, schedules collection at a draw station two blocks away for the following morning, and receives results in his patient portal within 48 hours. The results show an elevated HbA1c of 8.2%, prompting the pharmacist to recommend the patient contact his prescriber for a medication adjustment. The prescriber, in turn, increases the metformin dosage — a new prescription the patient fills at the same pharmacy. The platform captured a $65 transaction, the collection site earned a $30 routing fee, the reference laboratory processed three tests, and the pharmacy gained both a 9% revenue share and an incremental prescription fill. Every participant in the ecosystem benefited from a single pharmacist recommendation.

The geographic distribution of independent pharmacies is particularly valuable for platform network density. Independent pharmacies are disproportionately located in rural and medically underserved communities where other healthcare infrastructure is sparse. Over 40% of independent pharmacies serve communities of fewer than 20,000 people, and in many rural counties, the pharmacy is the only healthcare facility within a 30-minute drive. For the platform, onboarding pharmacies in these communities creates diagnostic access points in markets that would otherwise be unreachable without the capital cost of building physical sites. When combined with the telehealth layer and specimen routing to collection sites, the pharmacy channel enables the platform to offer a complete diagnostic pathway — from test recommendation through result delivery — in communities where no such pathway previously existed.

5. The Integrated Patient Journey

The power of the platform emerges not from any single participant, but from the seamless coordination of all six into a unified patient experience. Consider the following pathway — every participant operates independently, but the platform makes the experience feel integrated:

- **Step 1:** A consumer visits a collection site's branded e-commerce website (powered by the platform) and selects a comprehensive metabolic panel at a published cash price of $125.
- **Step 2:** The platform checks prescription authority requirements for the consumer's state. If required, the order is routed to a telehealth provider within the platform network for clinical authorization.
- **Step 3:** The consumer schedules specimen collection at the nearest collection site via the platform's scheduling engine, selecting a time slot from the site's real-time availability.
- **Step 4:** The specimen is collected and tracked via unique ID. The platform routes the specimen order to the appropriate reference laboratory via HL7 integration.
- **Step 5:** Results are received electronically from the laboratory, validated, and delivered to the patient via the branded portal with automated email/SMS notification.
- **Step 6:** If results indicate abnormality, the platform routes the patient to a telehealth consultation for interpretation. If imaging is indicated, the telehealth provider orders an MRI through the platform, routing the referral to a nearby imaging center at a published cash price of $450.
- **Step 7:** The imaging center receives the referral, schedules the patient, and delivers results back through the platform to the ordering clinician.

Total elapsed time: **48–72 hours**. Total cost to patient: **approximately $625** for a metabolic panel, telehealth consultation, and MRI. Through the traditional insurance-based pathway, the same sequence would take 4–6 weeks and cost $1,800–$4,000.

No single participant in this journey could deliver this experience alone. The collection site cannot offer telehealth. The telehealth provider cannot process specimens. The laboratory cannot schedule patients. The imaging center cannot receive electronic referrals from independent clinicians. **The platform makes them all work together.**

6. Market Sizing and Platform Economics

6.1 Total Addressable Market

The addressable population for cash-based diagnostic healthcare spans five distinct consumer demand segments, each representing significant and growing demand. On the supply side, the platform's six-sided network — collection sites, telehealth providers, laboratories, imaging centers, DPC practices, and independent pharmacies — provides the infrastructure to reach each segment through multiple access channels:

Segment	Population	Key Driver
Uninsured & Underinsured	92M+	No practical alternative
HDHP Holders (effectively self-pay)	85M+	Cash cheaper than deductible
Proactive Wellness Seekers	50M+	Wearable/self-directed health
Employer Population Health	Self-insured employers	Lower cost than traditional
Rural & Access-Constrained	60M+ in HPSAs	Telehealth + local collection

Conservative penetration modeling suggests that a platform connecting a planned network scaling to 3,000+ collection sites through channel partnerships, 200+ telehealth providers, 125+ reference laboratories, 300 imaging centers, 300 DPC practices, and 1,000 independent pharmacies could facilitate over **$1.5 billion in annual diagnostic GMV** by Year 5, with the platform capturing a combination of monthly access fees and transaction-based revenue share.

6.2 Platform Revenue Architecture

Unlike an operator model that captures the full consumer payment, a connectivity platform monetizes through infrastructure fees charged to each ecosystem participant. The six-sided network generates revenue from collection sites, telehealth providers, laboratories, imaging centers, DPC practices, and independent pharmacies through a combination of recurring access fees, transaction-based revenue share, and ecosystem routing fees:

- **Monthly Access Fees:** All six participant categories pay monthly access fees scaled to their feature set and value: collection sites ($499/month), telehealth providers ($750–$1,000/month), DPC practices ($500/month), independent pharmacies ($200/month), and imaging centers ($500–$1,000/month). Laboratory partners pay enterprise-level fees structured around their white-label agreements.
- **Transaction Revenue Share:** A percentage (typically 9%) of the consumer test price or imaging price flows to the platform as a revenue share on transactions facilitated through the ecosystem.
- **Collection Routing Fees:** When telehealth providers, laboratories, DPC practices, and pharmacies route specimen collection to network sites, the platform charges a $40 collection fee (of which $30 is paid to the collection site), generating $10 net margin per routed collection. Four distinct referral sources feed this flywheel, with pharmacies routing 100% and DPC practices routing 70–75% of their lab orders to collection sites.

- **White-Label Setup Fees:** Laboratories entering the ecosystem pay one-time setup and integration fees for e-commerce, branding, and test menu configuration.
- **Enterprise Platform Fees:** Large multi-site networks pay flat monthly platform fees for access across their entire organization.
- **DPC Practice Fees:** Direct primary care practices pay a $500/month access fee and 9% revenue share on laboratory and imaging GMV ordered through the platform. A one-time setup fee of $999 covers practice onboarding, ordering dashboard configuration, and collection site network mapping. DPC orders generate collection routing fees at a 70–75% routing rate, making each practice a consistent contributor to the collection fee flywheel.
- **Pharmacy Channel Fees:** Independent pharmacies pay a $200/month access fee and 9% revenue share on lab GMV, with no setup fee — the lowest entry point in the platform ecosystem. The pharmacy channel is uniquely efficient for the platform: 100% of pharmacy-originated lab orders route to collection sites (pharmacies do not perform phlebotomy), generating a $10 net collection fee on every transaction. This makes the pharmacy channel the single most productive driver of collection routing revenue per dollar of access fee collected.

6.3 Unit Economics Comparison: Operator vs. Platform

Metric	Operator Model	Platform Model
Revenue per $125 test	$125 (full price)	$11.25 (9% share)
Gross margin	50–70%	80%+
Capital to reach 3,000 sites	$50–100M+	$12–15M
Headcount at scale	2,000+	168
Regulatory burden	CLIA, state licenses, BAAs	Technology platform only
Revenue per employee (at scale)	~$250K	$987K
EBITDA margin at scale	45–55%	70.3%

The platform model trades top-line revenue per transaction for dramatically superior **capital efficiency, margin profile, and scalability. The platform operates as MAPL Health, Inc., a subsidiary of publicly traded ARCpoint Inc. (TSXV: ARC), providing listing and investor access from Day 1. Existing debt obligations have been restructured favorably: Cresso convertible note debt ($9M principal) converted to 29.5% equity stake, with a 20:1 share consolidation completed and multi-tranche financing structure in place. PubCo overhead is contained at $883K/year (95% OpCo / 5% PubCo allocation), preserving operating margin integrity.**. An operator serving 3,000 collection sites must manage 3,000 leases, thousands of employees, and complex multi-state regulatory compliance. A platform serving 3,000 collection sites manages 3,000 software subscriptions.

7. Cash-Pay Pricing: The Economic Foundation

7.1 Laboratory Testing

When laboratory services are decoupled from insurance billing, the economics transform dramatically. Reference laboratories publish separate self-pay price schedules typically 60–80% lower than insured rates. Emerging cash-pay lab networks offer even more aggressive pricing through volume commitments and technology-enabled efficiency.

Test	Insurance-Billed	Cash-Pay (Platform)
Comprehensive Metabolic Panel	$200–$400	$25–$45
Complete Blood Count	$100–$250	$15–$30
Lipid Panel	$150–$300	$20–$40
Thyroid Panel (TSH, T3, T4)	$200–$500	$35–$75
Testosterone Panel	$250–$600	$50–$100
STD Panel (comprehensive)	$400–$1,000	$99–$199

A comprehensive metabolic panel can be processed for **under $3 in variable cost at scale**. Even at consumer-facing cash prices of $25–$45, the margin structure supports the full value chain: reference lab processing, collection site operation, platform revenue share, and telehealth oversight.

7.2 Imaging Services

Modality	Hospital Price	Independent Cash-Pay	Savings
X-Ray	$200–$400	$75–$100	60–75%
Ultrasound	$300–$600	$120–$150	55–75%
CT Scan	$1,000–$3,000	$379–$395	60–87%
MRI	$1,200–$5,000	$250–$650	50–80%

The **$19 billion annual outpatient imaging market** represents significant platform revenue potential. With a blended average cash-pay price of approximately $325 and 6,000+ freestanding centers available as potential network participants, even modest penetration generates meaningful transaction volume for the ecosystem.

8. Technology Architecture

8.1 Core Platform Requirements

The technology infrastructure must accomplish several demanding objectives simultaneously: delivering a consumer-grade user experience, maintaining clinical-grade data security, integrating with a diverse ecosystem of laboratory instruments and health systems, and scaling efficiently from hundreds to thousands of connected participants.

- **Multi-Tenant Architecture:** Dynamic database connections per client with client-specific branding, configurations, and custom landing pages. Each collection site, laboratory, and telehealth provider operates under their own brand while sharing common platform infrastructure.
- **E-Commerce Ordering Engine:** Consumer-facing test ordering with transparent cash pricing, integrated scheduling, and payment processing via Stripe, Square, and QuickBooks.
- **Scheduling System:** Real-time appointment availability, time slot management, calendar views for collection sites, and multi-timezone support.
- **Laboratory Information System Integration:** Bidirectional HL7 messaging for electronic order transmission and result receipt. Support for HL7 FHIR, HL7 v2, and proprietary APIs to connect with partner laboratory platforms.
- **Result Management:** Lab result entry and validation, PDF result generation, automated notifications via email/SMS (Twilio), and patient portal access via magic-link authentication.
- **Specimen Tracking:** Unique specimen IDs from collection through accessioning and resulting, with full chain-of-custody visibility.
- **Telehealth Connectivity:** Interface connecting telehealth practitioners to the laboratory and imaging network, enabling electronic ordering, result viewing, and referral routing.
- **Revenue Cycle:** Cash payment processing, HSA/FSA acceptance, invoicing for B2B partners, and automated revenue share calculation and disbursement to platform participants.

8.2 Data Strategy

Longitudinal, integrated laboratory and clinical data is among the most valuable assets in healthcare. A platform that achieves meaningful scale will accumulate a dataset that is genuinely unique: laboratory results linked to telehealth-documented clinical presentations, collection site metadata, and patient-reported outcomes — all in a population actively engaged in their own health management.

This data asset, properly consented and de-identified, has substantial commercial value for pharmaceutical drug discovery, population health research, and value-based care analytics. A robust data governance framework — including transparent patient consent, validated de-identification methodology, and a data ethics committee — is both a compliance requirement and a competitive differentiator.

9. Health Equity and Social Impact

Critics of cash-pay healthcare models raise a legitimate concern: does eliminating insurance intermediaries risk creating a two-tiered system? The evidence suggests a more optimistic conclusion.

9.1 The Insurance Paradox

Insurance does not guarantee access. The United States has demonstrated that high insurance rates do not prevent large numbers of Americans from forgoing care due to cost. A system with genuinely transparent, dramatically lower cash prices — combined with sliding-scale programs and community health worker integration — can reach underserved communities more effectively than a nominally insured but practically inaccessible traditional system.

9.2 Platform-Enabled Equity Features

A platform architecture enables equity interventions that would be economically prohibitive for a single operator:

- **Network Density in Underserved Areas:** By connecting existing collection sites in HPSAs rather than building new facilities, the platform achieves coverage in medically underserved communities at marginal cost.
- **Transparent Sliding-Scale Pricing:** Platform-wide pricing rules enable income-verified discounts ensuring no patient pays more than 2–3% of monthly income for essential diagnostics.
- **Multilingual Platform Design:** Full translation support and culturally appropriate health communications in the top 15 languages spoken in target markets.
- **Telehealth as Equalizer:** For rural populations in access-constrained areas, telehealth combined with a nearby collection site may represent the most realistic pathway to diagnostic care.
- **Uncompensated Care Pools:** A percentage of platform revenue directed to a fund subsidizing care for patients below 200% of the federal poverty line.

10. Regulatory Landscape and Compliance Framework

A critical advantage of the platform model is its **dramatically reduced regulatory surface area** compared to an operator model. The platform does not operate laboratories (no CLIA requirements), does not employ clinicians (no medical practice licensing), and does not process specimens (no state laboratory permits). Regulatory compliance obligations reside primarily with the six categories of ecosystem participants — collection sites, telehealth providers, laboratories, imaging centers, DPC practices, and independent pharmacies — while the platform maintains compliance with technology, data, and payment requirements.

Key compliance domains for the platform include:

- **HIPAA Technical Safeguards:** The platform handles protected health information in transit and must maintain appropriate technical, administrative, and physical safeguards. Business Associate Agreements govern relationships with each connected participant.
- **State Telehealth Laws:** Telehealth providers operating through the platform must maintain compliance with state-specific prescribing authority, informed consent, and interstate licensing requirements. The platform facilitates compliance by routing orders only to appropriately licensed providers.
- **Direct Access Testing Laws:** State-by-state variation in whether consumers can order laboratory tests without a physician order. The platform's prescription authority integration automatically enforces state-specific requirements.
- **Payment Processing Compliance:** PCI-DSS compliance for cash payment processing, HSA/FSA payment acceptance, and automated invoicing.
- **Data Privacy:** State consumer privacy laws (CCPA, etc.), HIPAA Privacy Rule compliance, and transparent consent architecture for any data commercialization.
- **DPC Practice Integration:** DPC practices operate outside the insurance system, eliminating claims-based regulatory requirements. However, physicians ordering through the platform must maintain active state medical licenses and comply with state-specific laboratory ordering authority. The platform validates ordering physician credentials and routes orders only to appropriately credentialed providers. DPC-specific state registration requirements (applicable in approximately 30 states with DPC-enabling legislation) remain the responsibility of the individual practice.
- **Pharmacy Channel Compliance:** The platform's pharmacy model is designed to operate within existing pharmacy scope of practice in all 50 states. Pharmacists recommend diagnostic testing but do not order, collect specimens, or interpret results — activities that remain with physicians and laboratories within the ecosystem. No CLIA waiver is required at the pharmacy location because no specimen collection or testing occurs on-site. State board of pharmacy regulations governing patient counseling and medication therapy management apply to the pharmacist's recommendation activity, which falls within standard pharmacist practice authority. The platform maintains compliance documentation confirming that pharmacy-channel workflows do not trigger expanded scope-of-practice requirements.

11. Implementation Roadmap

Phase 1: Foundation (Months 1–12)

Build the core technology platform. Onboard initial collection site networks through channel partnerships with established aggregators such as i3screen and MOMs, which control 3,000–4,000+ sites (relationships established, formal agreements pending). As of March 2026, 118 collection sites are live on the platform. Launch telehealth provider connectivity in states with favorable direct access testing laws. Execute white-label partnership with first high-complexity laboratory partner. Establish proof of concept with at least one enterprise customer operating a multi-hundred-site network. Focus on validating unit economics and establishing the platform's value proposition to each participant category.

Phase 2: Market Penetration (Months 13–24)

Scale collection site network to 800+ active sites through channel partner activation. Expand telehealth provider network to 20+ practices. Onboard additional high-complexity laboratories through white-label program. Begin direct sales to freestanding imaging centers. Launch DPC practice onboarding targeting 50 initial practices and pharmacy channel activation targeting 100 independent pharmacies. Introduce collection fee routing as a cross-ecosystem revenue mechanism with four referral sources (telehealth, laboratories, DPC, and pharmacy). Achieve platform profitability on an EBITDA basis. Begin accumulating the longitudinal data asset. Financial modeling assumes multi-tranche funding through the public company structure (TSXV: ARC), with 20:1 share consolidation completed and interim, debt settlement, and growth financing tranches in place. Year 2 projections show $13.2M revenue and $2.9M EBITDA (21.6% margin), achieving platform profitability on an EBITDA basis. Sensitivity analysis demonstrates margin viability across the 45–70% range by Year 5. Conservative scenario assumes 85 collection sites operational in Year 1, with Year 5 conservative EBITDA of $41.1M at 48.4% margin. The platform currently has 118 collection sites live.

Phase 3: Scale and Optimization (Months 25–48)

Expand collection site network toward 3,000+ sites. Scale telehealth connectivity to 200+ providers across all 50 states via interstate compact licensing. Grow imaging center network to 300 sites through direct sales. Scale DPC practices to 300 with integrated lab and imaging ordering. Expand pharmacy network to 1,000 independent pharmacies, leveraging the channel as the primary driver of collection fee volume. Develop specialty care pathways (chronic disease management, men's and women's health, mental health) as high-value extensions. Evaluate data monetization opportunities with pharmaceutical and population health partners. Pursue geographic density in underserved markets through mobile collection, pharmacy-based access points, and enhanced home collection integrations.

Phase 4: Ecosystem Maturity (Year 5+)

At maturity, the platform is designed to facilitate over $1.5 billion in annual diagnostic and imaging GMV across a planned network scaling to 3,000+ collection sites through channel partnerships (118 live today), 200+ telehealth providers, 125+ laboratories, 300+ imaging centers, 300 DPC practices, and 1,000 independent pharmacies. Financial modeling projects Year 5 revenue of $165.7M at 93% gross margin, EBITDA of $116.6M (70.3% margin), and revenue per employee of $987K across a lean 168-person team. Conservative scenario

modeling yields $41.1M EBITDA at 48.4% margin. Network effects compound: each new participant makes the ecosystem more valuable to every existing participant. Collection sites gain more patient volume from telehealth, DPC, and pharmacy referrals. Telehealth providers gain access to more convenient collection points. Laboratories gain broader specimen sourcing. Imaging centers gain referral pipelines they could not build independently. DPC practices gain wholesale diagnostic pricing that strengthens their membership value proposition. Pharmacies gain a new revenue stream and incremental prescription volume driven by diagnostic findings.

The platform's strategic position at this stage is analogous to **Visa in payments**: it does not issue cards, operate banks, or make loans. It provides the infrastructure that allows independent financial institutions to interoperate. The healthcare connectivity platform does not operate labs, employ clinicians, or open collection sites. It provides the infrastructure that allows independent healthcare participants to function as an integrated ecosystem.

12. Conclusion

The healthcare system's fundamental dysfunction is not a mystery. It is the predictable outcome of a system that has interposed layers of administrative intermediaries between patients and providers, obscured prices, fragmented care, and designed itself around billing rather than healing.

Previous attempts to fix this have largely followed the same playbook: build a better vertically integrated healthcare company. Hire the clinicians. Open the sites. Own the patient. This approach requires massive capital, carries enormous operational risk, and competes directly with entrenched incumbents on their home turf.

The platform model takes a fundamentally different approach. It recognizes that the United States already has enough laboratories, enough collection sites, enough telehealth providers, enough imaging centers, enough DPC practices, and enough independent pharmacies to deliver excellent cash-based care to hundreds of millions of people. What it lacks is **the infrastructure to connect them**. The platform provides that connective tissue — enabling independent participants to coordinate seamlessly while retaining their autonomy, their brands, and their patient relationships.

The convergence of telehealth normalization, HDHP proliferation, laboratory technology advances, and consumer price consciousness has created conditions that did not exist a decade ago. A platform that connects the existing healthcare supply to the massive and growing cash-pay demand does not require a political revolution, a legislative supermajority, or a paradigm shift in physician training. It requires capital, technology, and the conviction that healthcare's biggest problem is not a shortage of providers — it is a shortage of connectivity.

— End of White Paper —

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